

Community Detection Report

Citation network community analysis

1. Dataset and scope

The analysis evaluates the detected communities in the citation network, including community sizes, coverage, modularity, and runtime.

2. Community-size statistics

Statistic	Value
Minimum community size	1
Maximum community size	109,371,240
Median	3
Mean	438,279.89
25th percentile	1
75th percentile	17.5
90th percentile	212.6
99th percentile	29,150

3. Coverage and modularity

Metric	Value
Number of communities	250
Singleton communities	91
Largest community fraction	99.8186%
Top 10 communities fraction	99.9865%
Node coverage	100%
Edge coverage	100%
Modularity	0.00002137
Analysis and modularity runtime	70,379.24 seconds (19.55 hours)

4. Interpretation

The community structure is highly imbalanced. One community contains nearly the entire network, while most communities are very small. The modularity is close to zero, indicating weak separation relative to the degree-preserving null model. The reported runtime corresponds to the statistics and modularity analysis; the original community-detection runtime should be reported separately if available in the detection logs.

5. Runtime note

The 19.55-hour value is the analysis and modularity-calculation runtime, not confirmed original detection runtime. Replace it with the original detection runtime if it is recorded in the community-detection logs.